Lab 5 - Negative Image Transformation

June 7, 2025

1 Theory

In digital image processing, the **negative transformation** is a fundamental operation that inverts the intensity values of an image. For a grayscale image with intensity levels in the range $[0, L-1]$, where L is typically 256 for 8-bit images, the negative of an image is obtained by subtracting each pixel value from the maximum intensity:

$$s = L - 1 - r$$

where:
- r is the original pixel value,
- s is the transformed (negative) pixel value,
- L is the number of possible intensity levels.

This operation effectively reverses the grayscale levels of the image, turning dark areas into light and vice versa. The negative transformation is particularly useful for enhancing details in dark regions of an image, making features that are difficult to see in the original image more visible.

[4]:

```
#import libraries
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt
```

[5]:

```
def Negative_Image(image):
  image = image.resize((400,400), Image.Resampling.LANCZOS)
  # convert to numpy array
  numpy_image = np.array(image)

  row = numpy_image.shape[0]
  column = numpy_image.shape[1]

  new_array = np.zeros(shape=(row,column))

  for i in range(row):
    for j in range(column):
      new_array[i][j] = 255 - numpy_image[i][j]

  negative_image = Image.fromarray(new_array)
```

```python
  negative_image = negative_image.convert("L")

  return negative_image
```

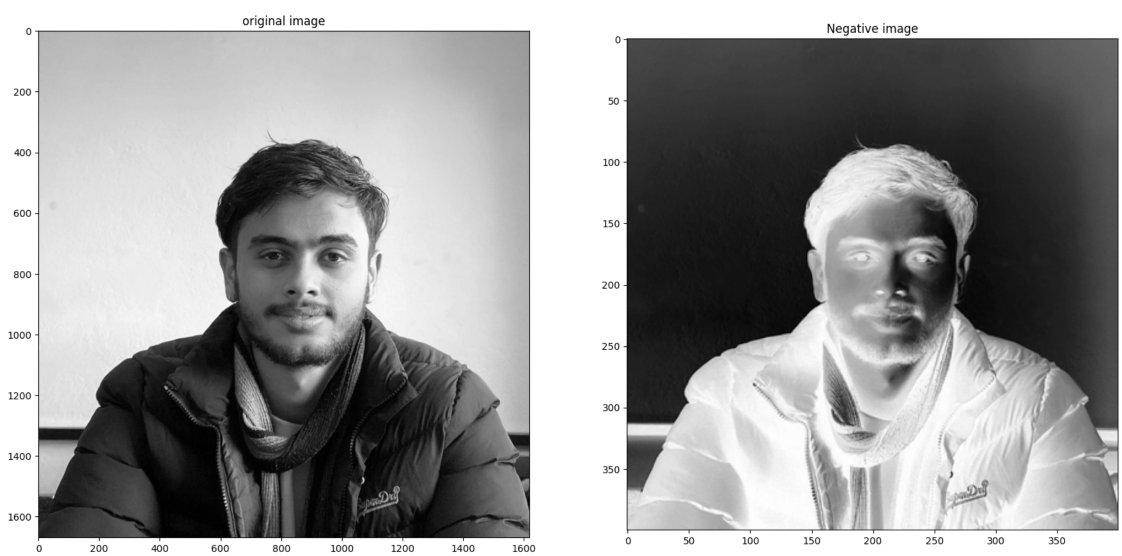
[6]:
```python
# reading image and converting to gray scale
image = Image.open('img/samir.jpg').convert('L')
# calling negative function
negative_image = Negative_Image(image)

#displaying the images
fig = plt.figure()
fig.set_figheight(20)
fig.set_figwidth(20)

fig.add_subplot(1,2,1)
plt.imshow(image, cmap='gray')
plt.title('original image')

fig.add_subplot(1,2,2)
plt.imshow(negative_image, cmap='gray')
plt.title('Negative image')
```

[6]: Text(0.5, 1.0, 'Negative image')



2 Conclusion

The negative image transformation is a simple yet powerful tool in digital image processing. By inverting the intensity values, it enhances features in dark regions and can reveal hidden details that are not easily visible in the original image. This technique is especially valuable in applications such as medical imaging, where subtle details in dark areas are critical for analysis. The experiment demonstrates how negative transformation can be implemented using Python and visualized using libraries such as PIL and Matplotlib.